



US012388220B2

(12) **United States Patent**
Matsumoto et al.

(10) **Patent No.:** **US 12,388,220 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **CONNECTOR**

(56) **References Cited**

(71) Applicant: **J.S.T. Mfg. Co., Ltd.**, Tokyo (JP)

U.S. PATENT DOCUMENTS

(72) Inventors: **Hiroyuki Matsumoto**, Yokohama (JP);
Shinya Masada, Yokohama (JP);
Hayato Chiba, Yokohama (JP)

5,370,550 A * 12/1994 Alwine H01R 13/641
439/352
6,824,417 B1 * 11/2004 Nimura H01R 13/641
439/352
8,926,355 B2 * 1/2015 Heil H01R 13/6275
439/352
11,456,553 B2 * 9/2022 Soh H01R 43/20
2017/0250492 A1 8/2017 Endo
2023/0335949 A1 * 10/2023 Matsumoto H01R 13/639

(73) Assignee: **J.S.T. MFG. CO., LTD.**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 329 days.

FOREIGN PATENT DOCUMENTS

(21) Appl. No.: **18/133,939**

JP 6213590 B2 10/2017

(22) Filed: **Apr. 12, 2023**

* cited by examiner

(65) **Prior Publication Data**

US 2023/0335949 A1 Oct. 19, 2023

Primary Examiner — Neil Abrams

(74) *Attorney, Agent, or Firm* — Kratz, Quintos &
Hanson, LLP

(30) **Foreign Application Priority Data**

Apr. 19, 2022 (JP) 2022-068594

(57) **ABSTRACT**

A connector includes: a connector body including a lever arm having a first lock portion locked to a to-be-locked member of a mating connector when the connector body is fitted to the mating connector; and a second lock portion provided on the connector body so as to be slidable in a width direction orthogonal to a fitting direction to the mating connector. When the connector body is fitted to the mating connector, the second lock portion is released from a locked state to the connector body, by the mating connector, is slid in the width direction to a position where the second lock portion interferes with the lever arm, and restricts a release operation of the first lock portion from the to-be-locked member.

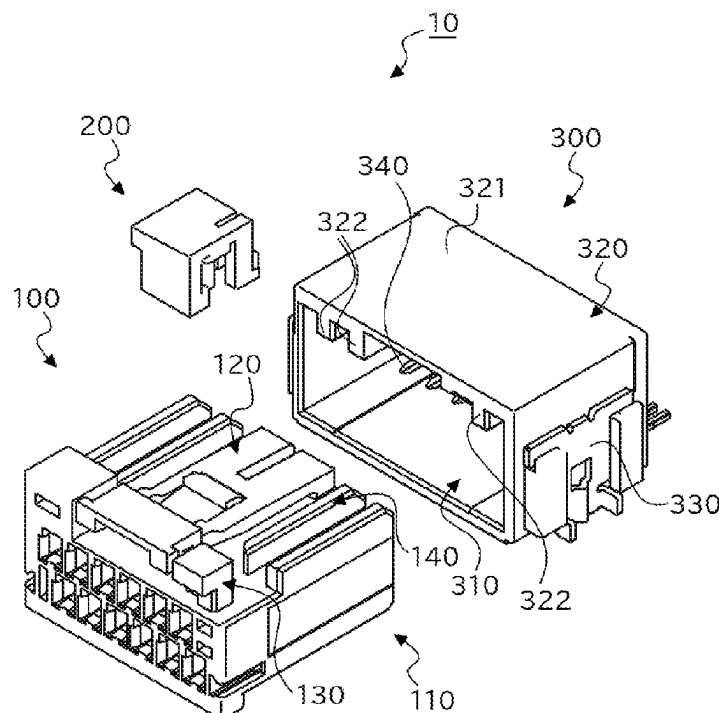
(51) **Int. Cl.**
H01R 13/639 (2006.01)

(52) **U.S. Cl.**
CPC **H01R 13/639** (2013.01)

(58) **Field of Classification Search**
CPC H01R 13/639; H01R 13/6272; H01R
13/6273

See application file for complete search history.

6 Claims, 8 Drawing Sheets



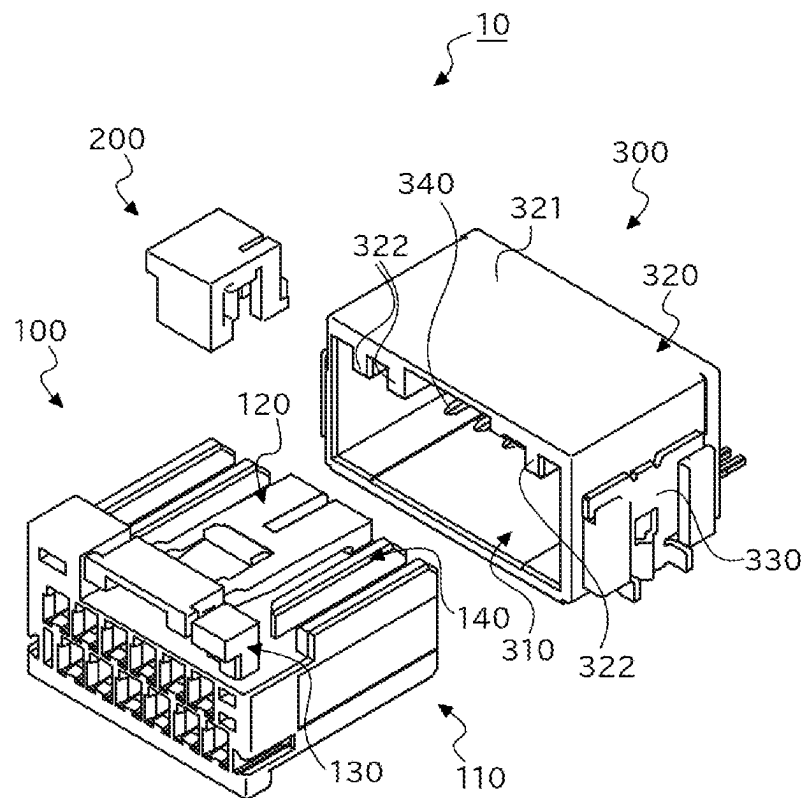


FIG. 1

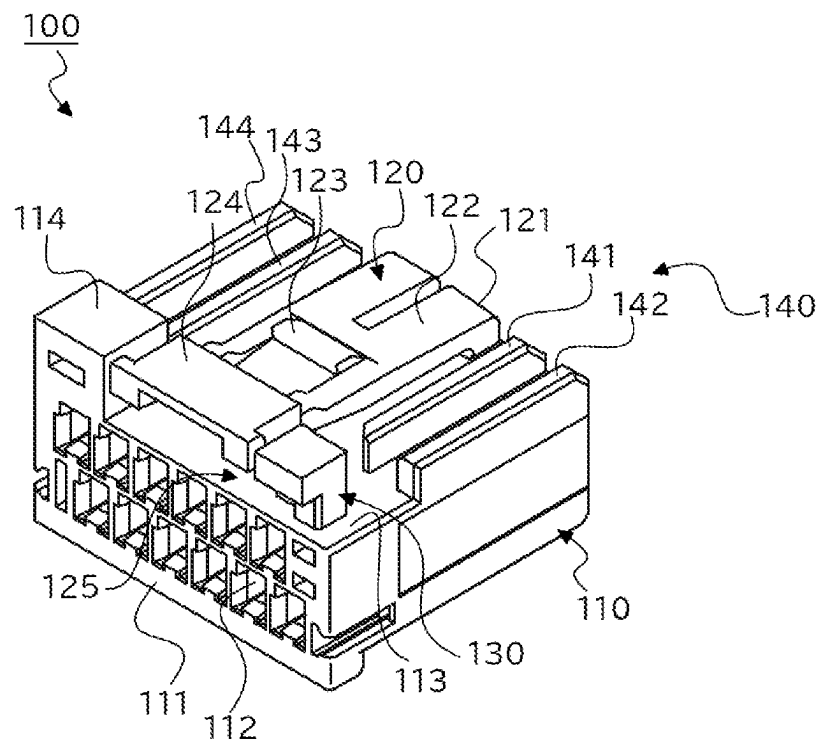


FIG. 2

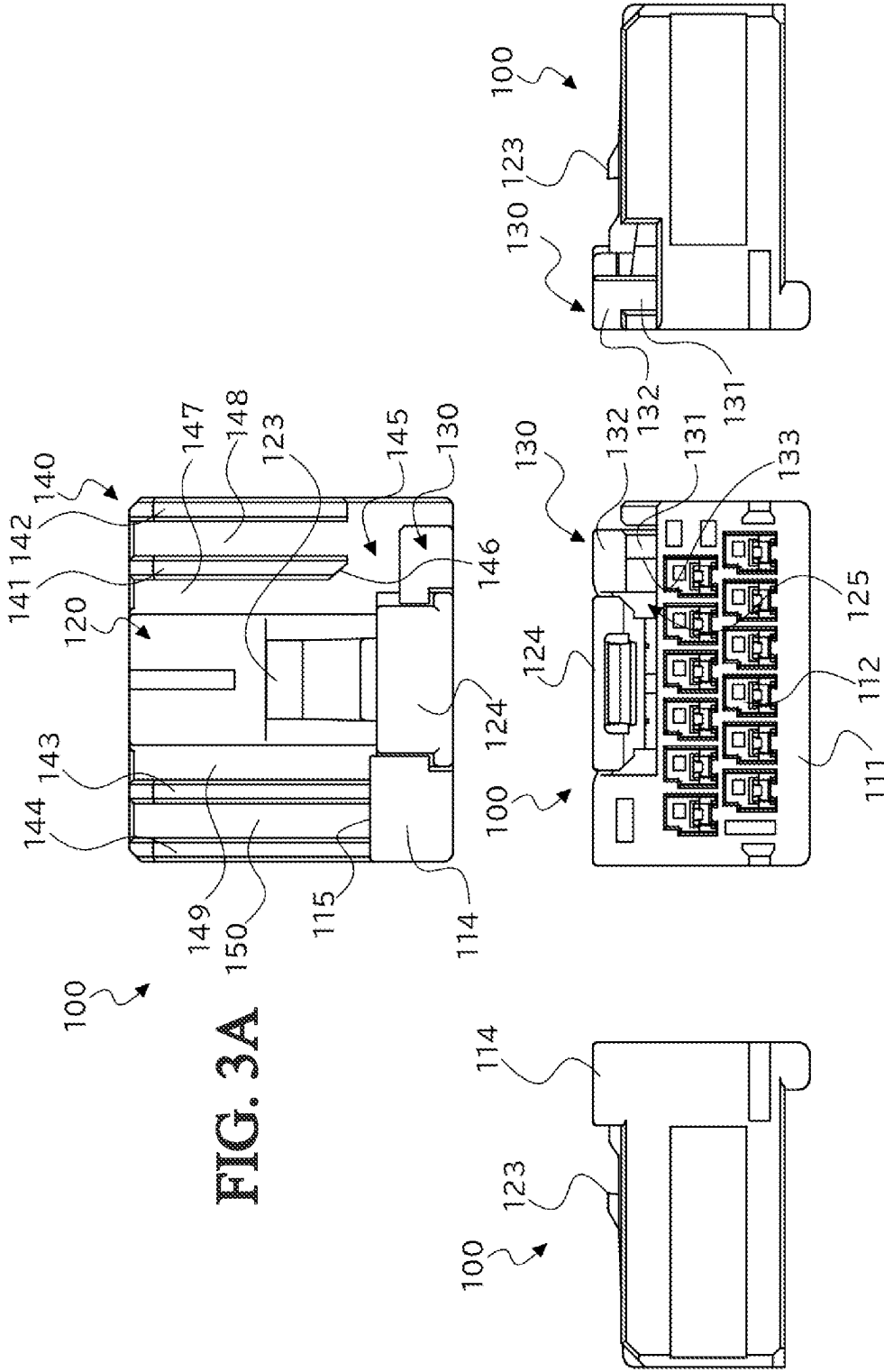


FIG. 3A

FIG. 3B

FIG. 3C

FIG. 3D

FIG. 4A

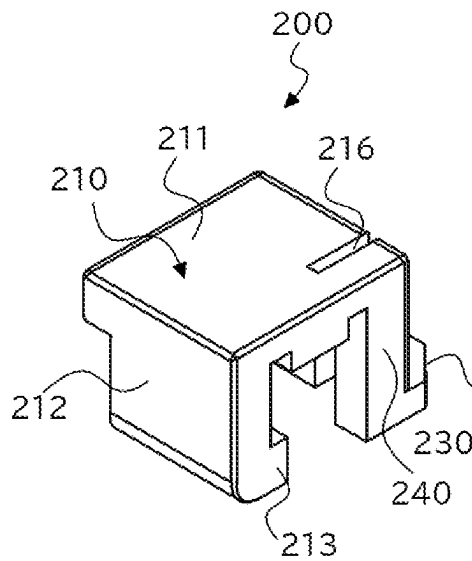


FIG. 4B

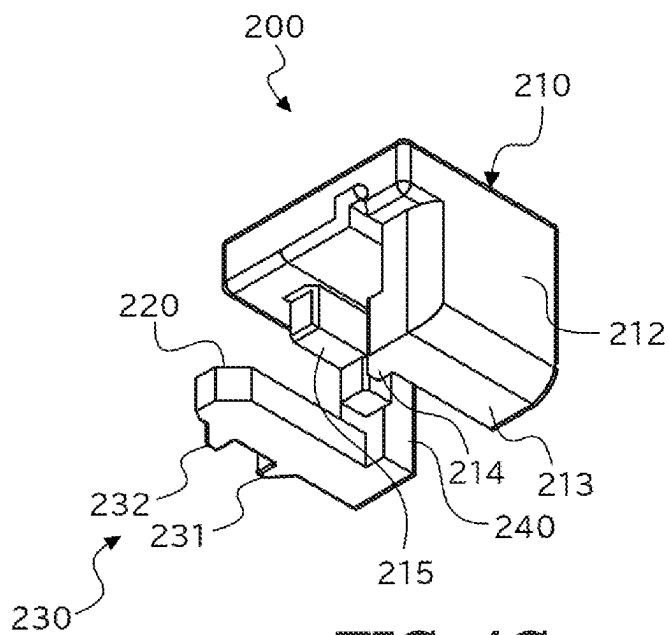
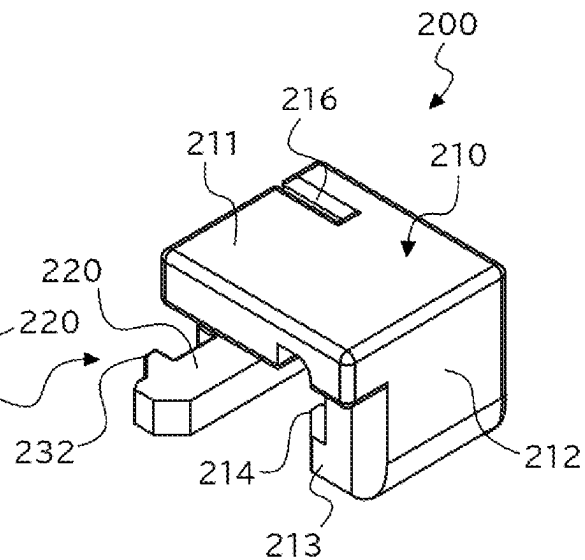
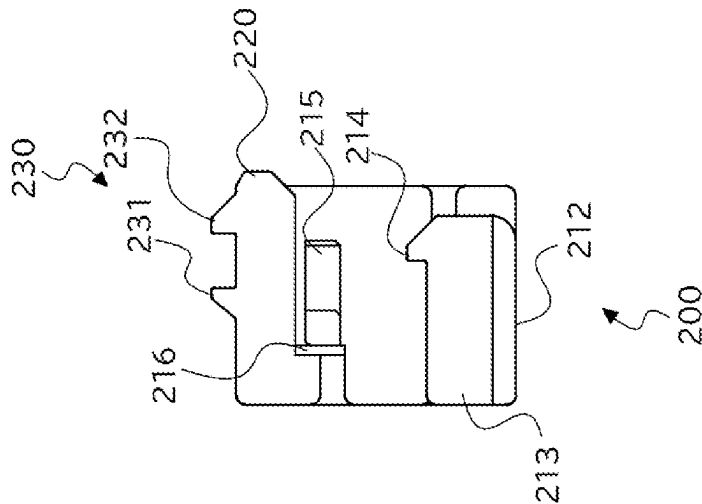
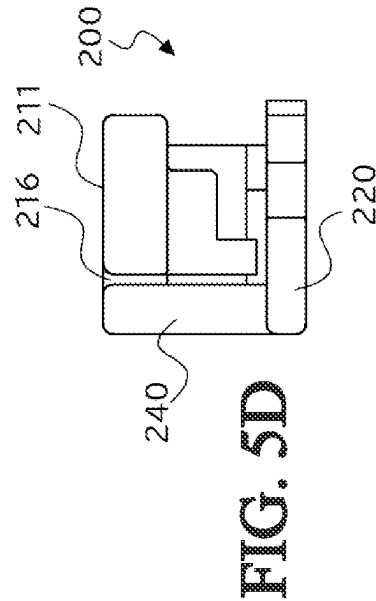
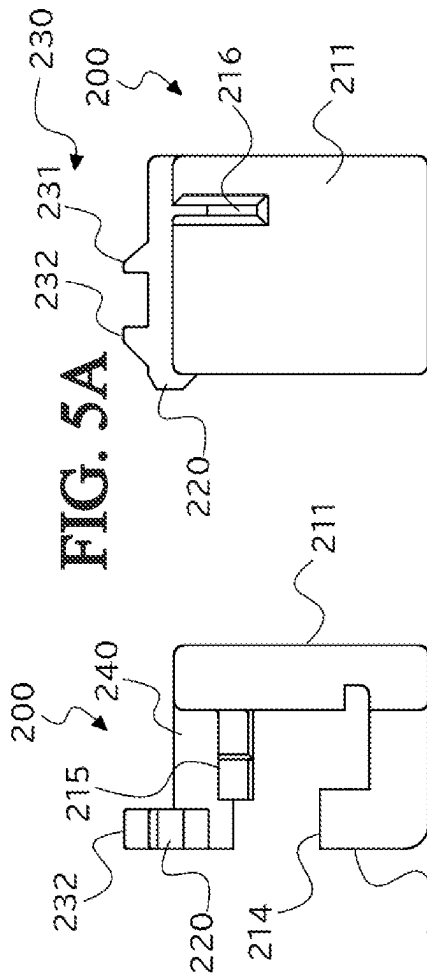
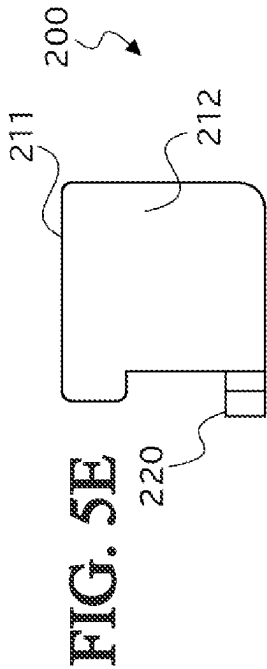
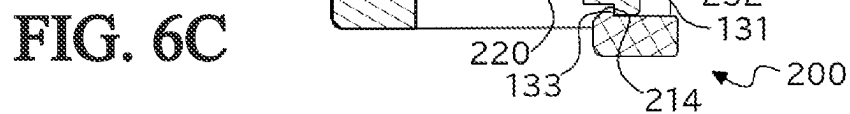
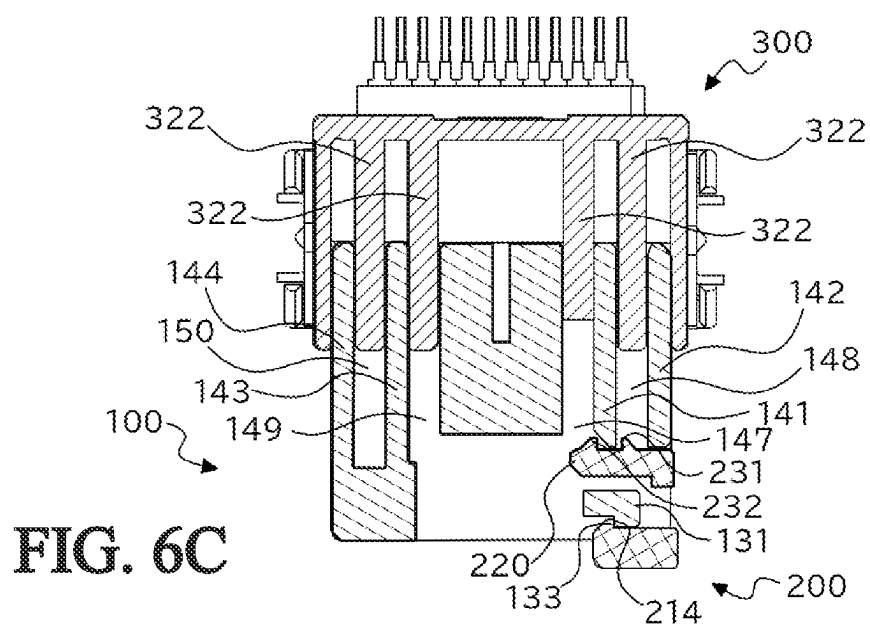
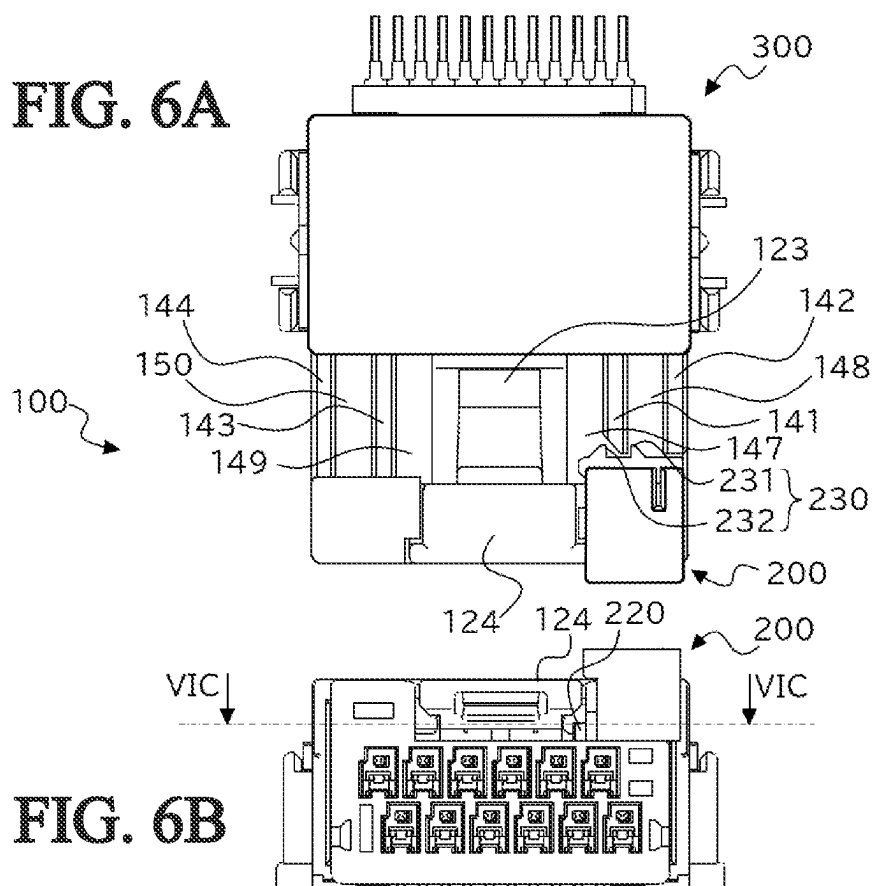
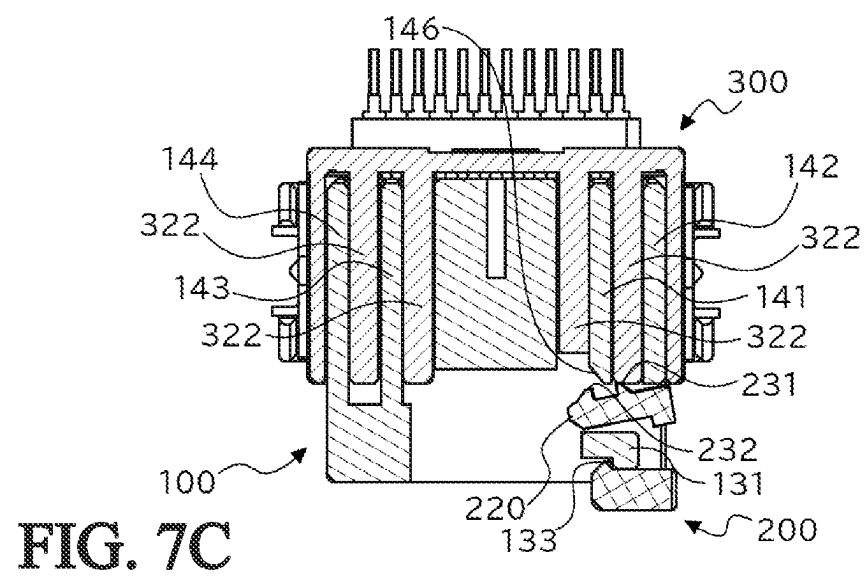
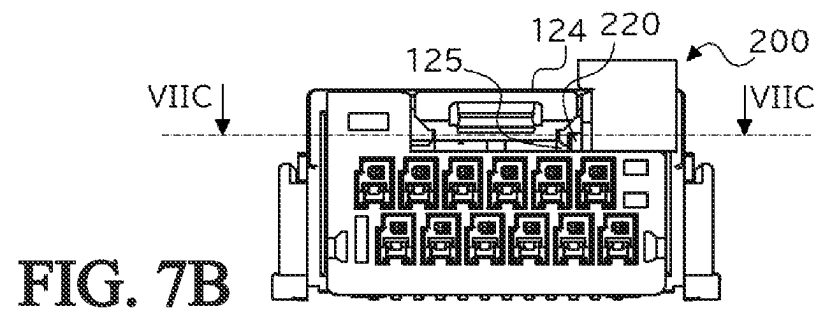
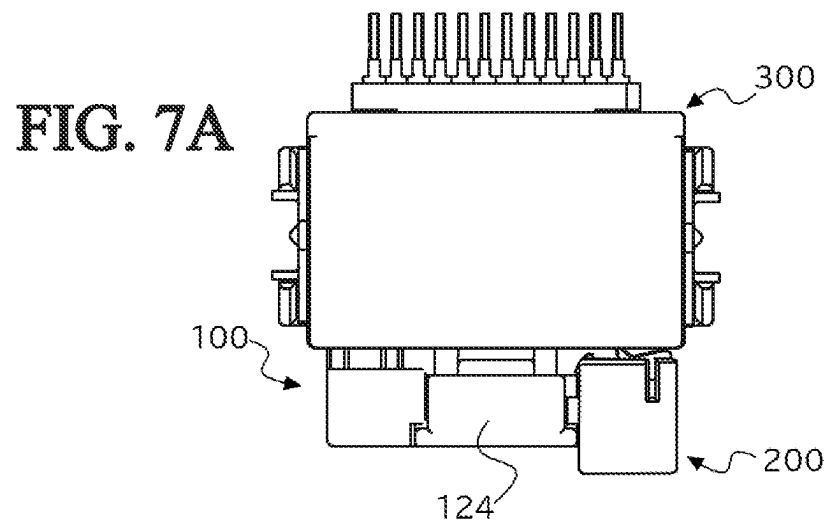
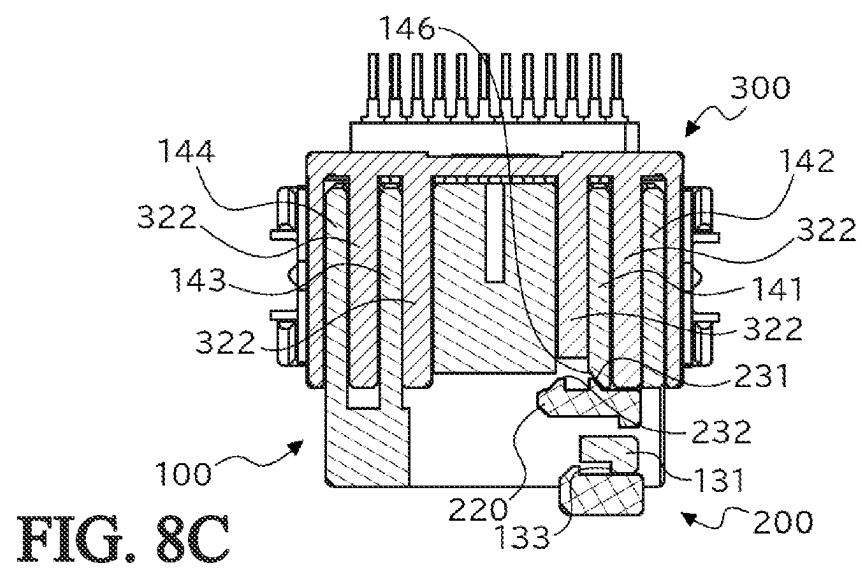
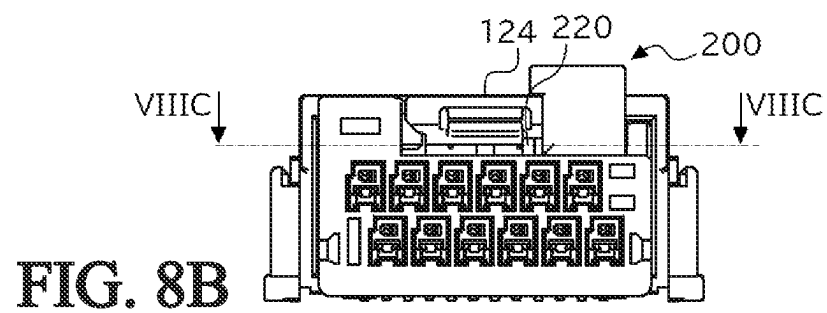
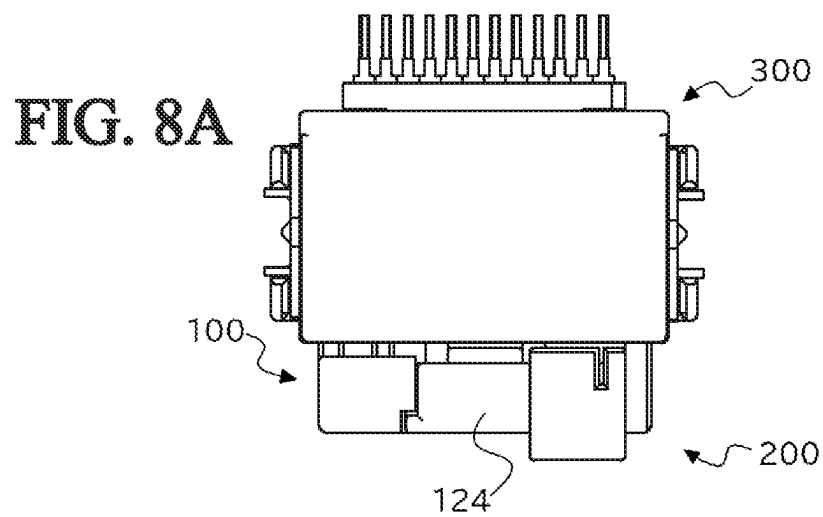


FIG. 4C









1

CONNECTOR**CROSS-REFERENCE TO RELATED APPLICATION**

This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-068594 filed on Apr. 19, 2022, the disclosure of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

The present invention relates to a connector, in particular, a connector including a lock portion that integrally holds connectors after the connectors are fitted to each other.

RELATED ART

Conventionally, in order to maintain a fitted state of connectors, there is a connector of a type in which a connector position assurance (CPA) member is moved in a connector fitting direction to be assembled to a connector body and operated.

For example, JP 6213590 B2 discloses a connector in which a slide member is inserted into a slide channel formed between an inner housing and a lock arm extending in a connector fitting direction in parallel with the connector fitting direction. In the connector, the inner housing is fitted to an outer housing, and the lock arm is locked to the outer housing. Then, the slide member restricts the push-down operation of the locked lock arm so as not to release the lock.

SUMMARY

However, in the connector position assurance mechanism configured as disclosed in JP 6213590 B2, the slide member needs to be arranged below the lock arm even at a temporary locking position, and there is a problem that the height reduction cannot be realized.

Therefore, an object of the present invention is to provide a connector having a structure capable of reducing the extension height of the lever arm with respect to the connector body and achieving the height reduction of the connector.

In order to solve the above problem, a connector according to the present invention includes:

a connector body including a lever arm having a first lock portion; and

a second lock portion provided on the connector body, wherein

the second lock portion is slidable in a width direction orthogonal to a fitting direction to a mating connector, the mating connector including a to-be-locked member, and

when the connector body is fitted to the mating connector, the first lock portion is locked to the to-be-locked member, and

the second lock portion is released from a locked state to the connector body, by the mating connector, and is slid in the width direction to a position where the second lock portion interferes with the lever arm, and is configured to restrict a release operation of the first lock portion from the to-be-locked member.

Furthermore, according to one aspect of the present invention, the lever arm extends in the fitting direction and has one side connected to the connector body and an other side forming an arm displacement space between the other

2

side and the connector body, and the second lock portion is placed on the connector body and configured to take two positions of a temporary locking position and a main locking position in the width direction, at the temporary locking position, the second lock portion is located outside the arm displacement space in the width direction, and at the main locking position, is at least partially inserted into the arm displacement space to restrict a release operation of the lever arm.

Furthermore, according to one aspect of the present invention, the connector body includes: a guide member extending in the width direction and configured to guide the second lock portion to move in the width direction; and a locking member, the second lock portion includes: a straddling seat portion that straddles the guide member; a claw member that is locked to the locking member at the temporary locking position; an arm restricting member that is inserted into the arm displacement space at the main locking position; and a connecting portion that elastically deformably connects the straddling seat portion and the claw member, and when the connector body is fitted to the mating connector, the claw member is released from a locked state to the locking member by being pushed in the fitting direction by the mating connector.

Furthermore, according to one aspect of the present invention, the guide member and the straddling seat portion form a falling-off prevention structure that prevents the second lock portion from falling off from the guide member.

Furthermore, according to one aspect of the present invention, the locking member is formed on the connector body as a part of a rib that extends in the fitting direction and guides fitting of the mating connector, and the claw member is locked to the part of the rib at the temporary locking position.

According to the above aspect of the present invention, when the connector body and the mating connector are fitted, the second lock portion is configured to move from the temporary locking position where the second lock portion does not interfere with the lever arm to the main locking position where the second lock portion interferes with the lever arm. Accordingly, the extension height of the lever arm with respect to the connector body can be reduced, and the height reduction of the connector can be achieved.

According to the above aspect of the present invention, at the temporary locking position, the second lock portion is not inserted into the arm displacement space, so that the height of the arm displacement space can be reduced.

According to the above aspect of the present invention, it is possible to release the locking between the claw member and the locking member only when the connector body is fitted to the mating connector.

According to the above aspect of the present invention, it is possible to prevent the second lock portion from falling off from the connector body.

According to the above aspect of the present invention, it is not necessary to form the locking member that locks the second lock portion to the connector body and the rib that guides the fitting of the mating connector as separate members.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is an exploded perspective view illustrating a configuration of a connector assembly according to an embodiment;

FIG. 2 is an external perspective view of a female connector according to the embodiment;

3

FIG. 3A is a top view of the female connector in FIG. 2, FIG. 3B is a rear view of the female connector, FIG. 3C is a right side view of the female connector, and FIG. 3D is a left side view of the female connector;

FIG. 4A is a perspective view of a CPA member when viewed from a rear upper side,

FIG. 4B is a perspective view of the CPA member when viewed from a rear upper side in a different direction from the case in FIG. 4A, and FIG. 4C is a perspective view of the CPA member when viewed from below;

FIG. 5A is a top view of the CPA member in FIGS. 4A to 4C, FIG. 5B is a right side view of the CPA member, FIG. 5C is a bottom view of the CPA member, FIG. 5D is a front view of the CPA member, and FIG. 5E is a rear view of the CPA member;

FIG. 6A is a top view of the connector assembly when the CPA member is at a temporary locking position in a course of fitting the female connector to a male connector, FIG. 6B is a rear view of the connector assembly in FIG. 6A, and FIG. 6C is a cross-sectional view taken along line VIC-VIC in FIG. 6B;

FIG. 7A is a top view of the connector assembly when the CPA member is in a state where it transitions from the temporary locking position to a main locking position in a state where the female connector and the male connector are fitted, FIG. 7B is a rear view of the connector assembly in FIG. 7A, and FIG. 7C is a cross-sectional view taken along line VIIC-VIIC in FIG. 7B; and

FIG. 8A is a top view of the connector assembly when the female connector and the male connector are completely fitted and the CPA member is at the main locking position, FIG. 8B is a rear view of the connector assembly in FIG. 8A, and FIG. 8C is a cross-sectional view taken along line VIIC-VIIC in FIG. 8B.

DETAILED DESCRIPTION OF THE INVENTION

A connector assembly 10 of the present embodiment will be described with reference to the drawings. Note that, in the embodiment described below, a female connector 100 of the connector assembly 10 is described as an example of the connector of the present invention, but the present invention is not limited to the female connector 100. The present invention should equally apply to other forms of connectors described in the claims.

As illustrated in FIG. 1, the connector assembly 10 includes the female connector 100, a connector position assurance member 200 (hereinafter, referred to as a CPA member 200) as a second lock portion of the present invention, and a male connector 300 as a mating connector of the present invention. The female connector 100 is inserted into a fitting opening 310 of the male connector 300, and is fitted to the male connector 300. The CPA member 200 is provided on a connector body of the female connector 100 so as to be slidable in a width direction orthogonal to a fitting direction to the male connector 300, and locks the two connectors so as not to be detached when the female connector 100 is completely fitted to the male connector 300.

Note that, in the following description, a direction in which the female connector 100 moves when being fitted to the male connector 300 is referred to as the fitting direction, a direction orthogonal to the fitting direction and in which the CPA member 200 moves is referred to as the width direction, and a direction orthogonal to both the fitting direction and the width direction is referred to as a vertical

4

direction. Furthermore, a side of the female connector 100 facing the male connector 300 in the fitting direction is referred to as “front”, and a side opposite to the side is referred to as “rear”.

As illustrated in FIG. 2, the female connector 100 includes a connector body housing 110 having a substantially rectangular parallelepiped shape. The connector body housing 110 includes a lever arm 120, a guide member 130, and ribs 140. The connector body housing 110 is made of resin, and is formed by, for example, injection molding. The connector body housing 110 has a rear end surface 111 in which a plurality of terminal insertion ports 112 into which female terminals (not illustrated) are inserted are formed, and the female terminals are electrically connected to terminal accommodating chambers formed therein.

The lever arm 120 has a front end 121 side connected to a front end of an upper surface 113 of the connector body housing 110, and extends rearward substantially in parallel with the upper surface 113. The lever arm 120 has an upper surface 122 in which a lock protrusion 123 as a first lock portion is formed so as to protrude upward in a center portion in the fitting direction. The lock protrusion 123 is locked to a to-be-locked member (not illustrated) of the male connector 300 when the female connector 100 is fitted to the male connector 300.

The lever arm 120 has a rear end side on which a locking release button 124 is formed. An arm displacement space 125 is formed between the locking release button 124 and the upper surface 113 of the connector body housing 110, and the locking release button 124 is movable in the vertical direction. In a case where the female connector 100 is fitted to the male connector 300, a user pushes the locking release button 124 downward. As a result, the locked state of the lock protrusion 123 to the to-be-locked member of the male connector 300 is released, and a state is brought about in which the female connector 100 can be removed from the male connector 300 by moving the female connector 100 rearward in the fitting direction.

The guide member 130 is formed on the upper surface 113 of the connector body housing 110 on one outer side of the locking release button 124 in the width direction. The guide member 130 extends in the width direction, and guides the CPA member 200 to move in the width direction. As illustrated in FIG. 3D, the guide member 130 has an L shape in a side view and includes a guide base portion 131 protruding upward from the upper surface 113 of the connector body housing 110 and a head portion 132 protruding rearward from the guide base portion 131. Moreover, as illustrated in FIG. 6C, the guide base portion 131 has a shape in which a side far from the locking release button 124 has a large thickness in the fitting direction, and a side close to the locking release button 124 has a small thickness in the fitting direction, and a boundary portion therebetween is a guide shoulder portion 133.

As illustrated in FIG. 3A, a plurality of ribs 140 are formed on the upper surface 113 of the connector body housing 110 so as to extend in the fitting direction in parallel with each other on both sides of the lever arm 120 in the width direction. These ribs 140 are for guiding the fitting of the male connector 300. The rib 140 is a ridge member formed so as to protrude upward from the upper surface 113 along the fitting direction, and in the present embodiment, when viewed from the rear surface side, a first rib 141 and a second rib 142 are formed on the right side (the left side as viewed from the front) of the lever arm 120 and a third rib 143 and a fourth rib 144 are formed on the left side.

5

The first rib **141** and the second rib **142** are formed from the front end portion of the upper surface **113** of the connector body housing **110** rearward to a point in front of the guide member **130** and the locking release button **124** in a side view. In particular, the rear end portion of the first rib **141**, which will be described later, functions as a locking member that holds the CPA member **200** at a temporary locking position. A CPA moving space **145** is formed between the first rib **141**, the second rib **142**, and the guide base portion **131** of the guide member **130**. The CPA moving space **145** has a distance enough to allow an arm restricting member, which will be described later, of the CPA member **200** to be interposed, deformed, and moved. The rear end portion of the first rib **141** has an inclined surface **146** in which a side facing the lever arm **120** is obliquely chamfered forward from the rear end. Furthermore, a first guide slot **147** is formed between the first rib **141** and the lever arm **120**, and a second guide slot **148** is formed between the first rib **141** and the second rib **142**.

The third rib **143** and the fourth rib **144** are formed from the front end portion of the upper surface **113** of the connector body housing **110** rearward up to a block-shaped portion **114** formed at the rear end portion of the upper surface **113** of the connector body housing **110**. A third guide slot **149** is formed between the third rib **143** and the lever arm **120**, and a fourth guide slot **150** is formed between the third rib **143** and the fourth rib **144**.

Next, the CPA member **200** will be described. As illustrated in FIGS. **4A** to **4C** and FIGS. **5A** to **5E**, the CPA member includes a straddling seat portion **210** straddling the guide member **130**, the arm restricting member **220**, claw members **230**, and a connecting portion **240**. The straddling seat portion **210** includes a top plate portion **211** spreading in the fitting direction and the width direction. A side plate portion **212** is provided substantially vertically downward from the rear end of the top plate portion **211**. The side plate portion spreads in the vertical direction and the width direction. Moreover, the side plate portion **212** is provided with a flange portion **213** extending frontward in the fitting direction from the lower end in parallel with the top plate portion **211**. One side end of the flange portion **213** in the width direction, that is, a side located on a lever arm **120** side when the CPA member **200** is attached to the guide member **130** is formed with a guide claw portion **214** protruding frontward in the fitting direction.

The lower surface of the top plate portion **211** is formed with an auxiliary guide protrusion **215** on the front side in the fitting direction so as to protrude substantially vertically downward in parallel with the side plate portion **212**. Therefore, the guide member **130** is fitted into an L-shaped space surrounded by the top plate portion **211**, the side plate portion **212**, the flange portion **213**, and the auxiliary guide protrusion **215**. As described above, the straddling seat portion **210** is attached so as to straddle the guide member **130**, and the CPA member **200** is movable in the width direction along the guide member **130**. In addition, in the straddling seat portion **210** and the guide member **130**, the flange portion **213** is prevented from moving upward by the head portion **132** of the guide member **130**, and the guide claw portion **214** is locked to the guide shoulder portion. This forms a falling-off prevention structure that prevents falling in upward and outward in the width direction.

The top plate portion **211** is formed with a slot **216** that extends in the fitting direction. A portion outside the slot **216** in the width direction forms the connecting portion **240** vertically extending downward from the top plate portion **211**. The connecting portion **240** is separated from other

6

portions of the top plate portion **211** and the auxiliary guide protrusion **215**, and can be displaced independently.

The connecting portion **240** has a lower end portion provided with the arm restricting member **220** extending in the width direction. The arm restricting member **220** protrudes from a side end portion of the top plate portion **211** in the width direction toward the center side in a top plan view. The arm restricting member **220** has a wall thickness that is about the same as or slightly smaller than the protruding height of the rib **140** of the connector body housing **110** from the upper surface **113** in the vertical direction. Furthermore, the claw members **230** that protrude frontward in the fitting direction and are locked to the first rib **141** when the CPA member **200** is at the temporary locking position are formed on a surface of the arm restricting member **220** on the front side in the fitting direction. The claw members **230** include a first claw member **231** provided on the outer side in the width direction and a second claw member **232** provided on the center side in the width direction. In other words, the connecting portion **240** elastically deformably connects the straddling seat portion **210** and the arm restricting member **220** on which the claw members **230** are formed.

Next, the male connector **300** as the mating connector will be described. As illustrated in FIG. **1**, the male connector **300** is formed by injection molding using resin or the like and includes a substantially box-shaped male housing **320** having the fitting opening **310** formed on the rear side in the fitting direction. The male housing **320** has a side surface in the width direction attached with a mounting fixture **330** for mounting a printed circuit board (not illustrated). Furthermore, the male housing **320** has a front surface in the fitting direction to which a plurality of male terminals **340** formed of a conductive material are fixed so as to protrude into the fitting opening **310**.

Moreover, the male housing **320** has an upper plate **321**, a lower surface of which is formed with a plurality of rib guides **322** protruding downward along the fitting direction. These rib guides **322** slide by being fitted into the above-described first guide slot **147** to the fourth guide slot **150**, and guide the insertion of the female connector **100** into the male housing **320**.

Next, assembly and operation of the connector assembly will be described. First, the CPA member **200** is placed on the connector body housing **110** of the female connector **100** so as to take two positions of the temporary locking position and a main locking position in the width direction. That is, the straddling seat portion **210** of the CPA member **200** is attached so as to straddle the guide member **130**. As particularly illustrated in FIG. **6C**, at the temporary locking position, the CPA member **200** is configured such that the guide claw portion **214** is locked to the guide shoulder portion **133** so that the CPA member **200** does not move outward in the width direction. Furthermore, the arm restricting member **220** is located in the CPA moving space **145**, and the first claw member **231** enters the second guide slot **148** between the first rib **141** and the second rib **142** and is locked to the rear end portion of the first rib **141**. As a result, the CPA member **200** takes the temporary locking position where the movement to the center side in the width direction is prevented. At the temporary locking position, the CPA member **200** is located outside the anti displacement space **125** in the width direction. That is, a distal end portion of the arm restricting member **220** on the center side in the width direction does not reach the arm displacement space **125** between the locking release button **124** and the upper

surface **113** of the connector body housing **110**, and the locking release button **124** is movable in the vertical direction.

Next, in a state where the CPA member **200** is attached, the connector body housing **110** of the female connector **100** is inserted into the fitting opening **310** of the male housing **320** of the male connector **300**. At this time of operation, the rib guides **322** formed on the lower surface of the upper plate **321** of the male housing **320** slide by being fitted into the first guide slot **147** to the fourth guide slot **150**, and guide the insertion of the connector body housing **110** of the female connector **100** into the male housing **320**.

Next, as illustrated in FIGS. 7A to 7C, the female connector **100** is inserted into the male housing **320** of the male connector **300** toward the front side in the fitting direction to establish a fitted state. At this time of operation, the lock protrusion **123** of the lever arm **120** is locked to the to-be-locked member (not illustrated) formed on the lower surface of the upper plate **321** of the male connector **300**. Furthermore, a state is brought about in which the female terminals (not illustrated) and the male terminals **340** are connected to each other and electrically conductive to each other.

Furthermore, at this time of operation, the CPA member **200** is mounted such that the straddling seat portion **210** straddles the guide member **130**, and thus, cannot move frontward and rearward in the fitting direction. On the other hand, the connecting portion **240** and the arm restricting member **220** can be displaced with a connecting portion between the connecting portion **240** and the top plate portion **211** as a fulcrum by the slot **216**. Accordingly, as particularly illustrated in FIG. 7C, a rear end of the rib guide **322** of the male connector **300** comes into contact with the first claw member **231**, and the first claw member **231** is pushed rearward in the fitting direction. Then, the first claw member **231** moves rearward in the fitting direction of the second guide slot **148**. As a result, the distal end portion of the arm restricting member **220** on the center side in the width direction is displaced rearward in the fitting direction, and the locked state of the first claw member **231** to the first rib **141** is released. At this time of operation, a state is brought about in which the CPA member **200** is movable toward the center side in the width direction. That is, the CPA member **200** is released from the locked state to the connector body housing **110** by the fitting of the male connector **300**.

Next, as illustrated in FIGS. 8A to 8C, the CPA member **200** is slid toward the center side in the width direction to the main locking position where the CPA member **200** interferes with the locking release button **124** of the lever arm **120**. That is, when the CPA member **200** is slid toward the center side in the width direction, the first claw member **231** gets over the first rib **141**, displacement of the arm restricting member **220** returns to the original state, and the first claw member **231** enters the first guide slot **147**. Then, the distal end portion of the arm restricting member **220** on the center side in the width direction reaches the arm displacement space **125** between the locking release button **124** and the upper surface **113** of the connector body housing **110**, and a state is brought about in which the locking release button **124** cannot move downward. That is, in the CPA member **200** at the main locking position, the arm restricting member **220** is inserted into the arm displacement space **125**. As a result, the lever arm **120** cannot move in the vertical direction inside the fitting opening **310** of the male housing **320**, and the operation of releasing the lock protrusion **123**

from the to-be-locked member formed on the lower surface of the upper plate **321** of the male connector **300** is restricted.

When releasing the fitting between the female connector **100** and the male connector **300**, first, the CPA member **200** is moved outward in the width direction. As a result, the first claw member **231** of the CPA member **200** comes into contact with the inclined surface **146** of the first rib **141**, so that the first claw member **231** is pushed rearward in the fitting direction, and a state is brought about in which the arm restricting member **220** is moved rearward in the fitting direction. Furthermore, at this time of operation, a state is brought about in which the distal end portion of the arm restricting member **220** on the center side in the width direction is retracted outward in the width direction from the arm displacement space **125** between the locking release button **124** and the upper surface **113** of the connector body housing **110**, and a state is brought about in which the locking release button **124** is movable downward. As a result, the lever arm **120** can move downward inside the fitting opening **310** of the male housing **320**, and the lock protrusion **123** can be released from the to-be-locked member formed on the lower surface of the upper plate **321** of the male connector **300**. The first claw member **231** gets over the rear end of the first rib **141** and moves to the rear of the second guide slot **148**, but the rib guide **322** is interposed in the second guide slot **148**, so that a state is brought about in which the first claw member **231** is pushed rearward of the rib guide **322**.

In this state, by pushing the locking release button **124** downward to release the lock protrusion **123** from the to-be-locked member of the male connector **300** and pulling out the female connector **100** from the fitting opening **310** of the male connector **300** rearward in the fitting direction, the female connector **100** can be removed from the male connector **300**.

Furthermore, at this time of operation, displacement of the arm restricting member **220** to the rear side in the fitting direction returns to the original state, and the first claw member **231** enters the second guide slot **148** and is locked to the rear end portion of the first rib **141**. Furthermore, the arm restricting member **220** is located in the CPA moving space **145**. As a result, the CPA member **200** is brought into a temporarily locked state in which the movement to the center side in the width direction is prevented. Furthermore, the guide claw portion **214** is locked to the guide shoulder portion **133**, so that the CPA member **200** is brought into a state in which the CPA member **200** does not move outward in the width direction.

The configuration and operation of the connector assembly **10** according to the present embodiment have been described heretofore. Note that, in the above embodiment, a case has been described where the rear end portion of the first rib **141** formed on the connector body housing **110** is used as the locking member, and the first claw member **231** of the CPA member **200** is locked thereto at the temporary locking position. However, the present invention is not limited thereto. The rear end portion of any one of the second rib **142** to the fourth rib **144** may be used as the locking member, and any one of the claw members formed in the CPA member **200** may be locked thereto.

Furthermore, in the above embodiment, a case has been described where the guide member **130** is arranged on the upper surface **113** of the connector body housing **110** on the right side as viewed from the rear (the left side as viewed from the front) in the fitting direction with respect to the lever arm **120**, and the CPA member **200** is arranged thereon. However, the present invention is not limited thereto, and

the guide member **130** and the CPA member **200** may be provided on the left side as viewed from the rear in the fitting direction. Furthermore, the lever arm **120**, the guide member **130**, and the CPA member **200** may be provided on the lower surface of the connector body housing **110**.

Furthermore, in the above embodiment, a case has been described where two ribs are formed on each side of the lever arm **120** in the width direction, but the present invention is not limited thereto. It is sufficient that at least one rib is formed on a side where the CPA member **200** is arranged, and two, three, or more ribs may be formed. Furthermore, ribs, the numbers of which are different on the left and right in the width direction, may be formed. Furthermore, the intervals between the ribs may be different on the right and left. In this case, if rib guides, the number and width of which are corresponding to the ribs, are formed in the male connector **300**, it is also possible to impart a misconnection identification function of identifying the connection of different types of connectors.

What is claimed is:

1. A connector comprising:

a connector body including a lever arm having a first lock portion; and

a second lock portion provided on the connector body, wherein

the second lock portion is slidable in a width direction orthogonal to a fitting direction to a mating connector, the mating connector including a to-be-locked member, and

when the connector body is fitted to the mating connector, the first lock portion is locked to the to-be-locked member, and

the second lock portion is released from a locked state to the connector body, by the mating connector, and is slid in the width direction to a position where the second lock portion interferes with the lever arm, and is configured to restrict a release operation of the first lock portion from the to-be-locked member.

2. The connector according to claim 1, wherein the lever arm extends in the fitting direction and has one side con-

nected to the connector body and an other side forming an arm displacement space between the other side and the connector body, and

the second lock portion is placed on the connector body and configured to take two positions of a temporary locking position and a main locking position in the width direction, at the temporary locking position, the second lock portion is located outside the arm displacement space in the width direction, and at the main locking position, is at least partially inserted into the arm displacement space to restrict a release operation of the lever arm.

3. The connector according to claim 2, wherein the connector body includes: a guide member extending in the width direction and configured to guide the second lock portion to move in the width direction; and a locking member,

the second lock portion includes: a straddling seat portion that straddles the guide member; a claw member that is locked to the locking member at the temporary locking position; an arm restricting member that is inserted into the arm displacement space at the main locking position; and a connecting portion that elastically deformably connects the straddling seat portion and the claw member, and

when the connector body is fitted to the mating connector, the claw member is released from a locked state to the locking member by being pushed in the fitting direction by the mating connector.

4. The connector according to claim 3, wherein the guide member and the straddling seat portion form a falling-off prevention structure that prevents the second lock portion from falling off from the guide member.

5. The connector according to claim 3, wherein the locking member is formed on the connector body as a part of a rib that extends in the fitting direction and guides fitting of the mating connector, and the claw member is locked to the part of the rib at the temporary locking position.

6. The connector according to claim 4, wherein the locking member is formed on the connector body as a part of a rib that extends in the fitting direction and guides fitting of the mating connector, and the claw member is locked to the part of the rib at the temporary locking position.

* * * * *